

Policy incentives, government subsidies, and technological innovation in new energy vehicle enterprises: Evidence from China

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ABSTRACT

As the first step toward establishing the development of new energy vehicles (NEVs) as a national strategic priority, China's New Energy Vehicle Pilot City (NEVPC) policy has far-reaching significance for helping to develop China's capabilities in NEV industry. The existing literature focuses on the city- and industry-level impacts of this policy but ignores the important role of individual enterprises as innovation agents. In addition, in assessing the policy's effects technological innovation for NEV enterprises is typically measured in terms of output quantity, with no emphasis on innovation efficiency. This study uses balanced panel data from 2006 to 2020 covering 196 listed Chinese NEV enterprises to examine the mechanism and impact of the NEVPC policy on technological innovation within NEV enterprises. We show that the policy has achieved the goal of expanding government subsidies and had a significantly positive impact on the number of technological innovations produced by these enterprises, via a partial mediation effect. However, neither the policy nor the mechanism of government subsidies is effective in improving the efficiency of technological innovations for NEV enterprises. These results have important implications for formulating and improving policies for the NEV industry.

1. Introduction

Developing a new energy vehicle industry (NEV) is important in addressing climate change and the global energy crisis (Gass et al., 2014). As part of a new round of global technological innovations, the NEV industry has emerged as strategically important in accelerating climate change-related innovation in countries around the world (Meckling and Nahm, 2019). As it is still in its early stages of the development, the industry is facing the dual pressure of low-cost competition from traditional vehicles (internal combustion engines) and the high cost of renewable energy. It may be challenging for the industry to address these pressures through the efforts of NEV enterprises alone (Siddiqui et al., 2007)—governments in countries that are implementing preferential policies to support NEVs can help the industry to flourish.

China is one of the largest, most economically important developing countries in the world, and the government regards the growth of the NEV industry as an essential means of reducing energy use, carbon emissions, and smog (Zhang and Qin, 2018). The promotion and development of NEVs in China has expanded rapidly in recent years (Liu et al., 2020a; Yuan et al., 2015). The New Energy Vehicle Pilot City

(NEVPC) policy launched in 2009 is a landmark move by the Chinese government to elevate the strategic importance of the country's NEV industry at a national level. This government subsidy policy for China's NEV industry has now gone through two full five-year planning periods. As a market orientation toward NEVs has developed, the promotional role played by industrial policies has gradually diminished and the focus has shifted to technological support (Dong and Liu, 2020). The key to developing NEVs lies in the core technology, and technological innovation is an important factor in productivity and competitiveness. It is important to understand the impact this NEV subsidy policy has had on innovation activities among NEV enterprises, and whether the effects of the policy have been significant.

The existing literature focuses on the effects of this policy at the city and industry levels but ignores the important role of individual enterprises as innovation agents. Wang et al. (2021) studied the impact of the NEVPC policy on urban green innovation and emission reduction and found that it significantly reduces CO₂ emissions in pilot cities but also significantly inhibits green innovation. Liu et al. (2020b) found that the NEVPC policy increases electric vehicles sales in pilot cities, especially in cities with a high demand for NEVs. However, in measuring the policy's effects, assessing technological innovation is primarily limited to the

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number of technological innovations, not innovation efficiency. For example, Zhao et al. (2019b) used a propensity score matching (PSM) and difference-in-differences (DID) model and showed that the NEVPC policy helped to increase the number of patent applications by NEV manufacturers.

In addition, the NEVPC policy supports the NEV industry primarily through government subsidies. However, studies show that the effect of these subsidies on enterprises' technological innovation is debatable. One view holds that, based on the spillover effect of public goods, government subsidies can effectively alleviate positive externalities of innovation activities and market failures caused by inevitable knowledge leakage (Choi and Lee, 2017). The non-reimbursable nature of government subsidies significantly reduces R&D costs and lowers the risk associated with innovation activities (Huang et al., 2019), which in turn increases the likelihood that enterprises will pursue technological innovation (Bronzini and Piselli, 2016). The other opposing view is that government subsidies are detrimental to enterprise-level innovation, are treated as a partial substitute for internal R&D expenditures (Ahn et al., 2020), and inhibit enterprise innovation performance by crowding out R&D investment (Yu et al., 2016). This leads us to ask whether the NEVPC policy has effectively achieved the government's objectives by providing financial subsidies for the NEV industry. Have government subsidies, as the core policy measure, contributed to the NEV industry's innovative activities? To date, there are no clear answers to these questions.

Therefore, this study analyzes the influence relationship between the NEVPC policy and technological innovations of NEV enterprises. The study contributes to the literature as follows: (1) Using the DID method, we quantitatively evaluate the impact of the NEVPC policy on two dimensions of technological innovation for NEV enterprises. This approach enriches the theoretical understanding of how external policy incentives influence the innovation ecosystem of the NEV industry; (2) From the perspective of government subsidies, we analyze and verify the impact mechanism of the NEVPC policy on technological innovation within NEV enterprises; (3) To better understand technological innovation efficiency, we use the data envelopment analysis window method to measure changes in technological innovation efficiency in NEV enterprises. The goal is to analyze the impact of the NEVPC policy incentives on the efficiency of technological innovation within NEV enterprises.

The remainder of this study is organized as follows. Section 2 introduces the background of the NEVPC policy. Section 3 defines the sample, methodology, and variables used in the empirical analysis. Section 4 presents the empirical results that identify causality between the policy effect and technological innovation of NEV enterprises. Section 5 concludes.

2. Policy background

In response to pressures to combat climate change and shortages of fossil fuel-based energy the global vehicle manufacturing sector is undergoing a transformation focused on developing NEVs. In recent years, the world's major vehicle enterprises have pursued NEV innovation as essential to achieving their long-term development goals. In addition, the countries that are home to the world's leading vehicle companies are strengthening policies to support and improving strategies to develop their NEV industries. Developing NEV industries is of great importance as it can affect a country's national energy security, energy conservation, emission reduction, technological prowess, and overall economic development.

The Chinese government's support for the NEV industry can be traced back to critical technical projects for electric vehicles covered in the "Eighth Five-Year Plan" period. For more than 30 years, the Chinese government's innovation policy for the NEV industry has covered various aspects of emerging industries ranging from technology research and development (Zuo et al., 2019) to commercialization (Zhao et al.,

2019a). Beginning in 2004, a series of policies focused on structural adjustments to the vehicle industry as well as energy conservation and emission reduction encouraged enterprises to expand the production of NEVs. The 2008 Beijing Olympic Demonstration Project produced the first large-scale demonstration of China's electric vehicle industry, with a total of 595 electric vehicles providing transportation during the Olympics. Nonetheless, the size of China's NEV market is still insignificant, and the critical bottleneck is that the immature technology of NEVs, in particular battery performance, vehicle economy, and safety, cannot meet consumers' general requirements. In short, development of the NEV industry still faces multiple challenges, including technical route selection, infrastructure construction, and consumer market guidance.

In 2009, in response to the many challenges facing the NEV industry the State Council of China issued the "Vehicle Industry Adjustment and Revitalization Plan," which for the first time raised the development of NEVs to a national strategic priority and proposed the NEVPC policy. As one of the measures in the plan, the Ministry of Finance and the Ministry of Science and Technology jointly issued the "Notice on Carrying out the Pilot Work of Demonstration and Promotion of Energy-Saving and NEVs," designating 13 cities, including Beijing, Shanghai, Changsha, and Hangzhou, to carry out pilot projects to demonstrate this effort. On May 31, 2010, five ministries of China jointly issued the "Notice on Expanding the Demonstration and Promotion of Energy Conservation and NEVs in the Public Service Field," naming seven other pilot cities, including Tianjin and Guangzhou, as the second group of NEV pilot cities. In June of the same year, five more cities, including Shenyang and Hohhot, were approved as the third group of NEV pilot cities. These 25 pilot cities have unique advantages in developing energy-saving technologies and NEVs, given their strengths in various modes of transportation, developing energy-saving technologies for NEVs, and related industries. For example, Tianjin, a municipality directly under the central government, ranks among the top in terms of its number of buses. Not only does Tianjin Public Transport have a one hundred-year history of operations, compared with other cities its public transport system is relatively complete, providing a good foundation for developing energy efficiencies and NEVs for public services. Guangzhou was one of the first cities in China to operate hybrid electric vehicles; as early as January 2008, Guangzhou had 30 hybrid electric vehicles in commercial operation.

The documents issued by the government ministries clearly state that the central government would give these 25 pilot cities a one-time fixed subsidy to purchase energy-saving NEVs such as hybrid electric vehicles, pure electric vehicles, and fuel cells, taking into account factors such as scale effects and technological progress. At the same time, the local governments had to arrange funds to provide appropriate financial subsidies to construct and maintain facilities to support energy-saving technologies and NEVs and to ensure the pilot work would progress by improving the urban infrastructure. After the pilot policy was introduced, each pilot city took action. In a short period, more than 10 NEV industry alliances and industrial bases were established across the country, providing necessary support in selecting technical routes for the development of NEVs.

China's NEVPC policy uses financial subsidies for the 25 pilot cities as its primary policy tool. The policy's goals are to explore the technological paths, business models, and development strategies of this emerging industry. By supporting technological innovation within NEV enterprises, government subsidies help to create standards for implementing new technology and investing in equipment for those enterprises, and promote R&D efforts to improve the use of various technology resources. New technologies and increased investment in equipment can also promote quality improvements in an enterprise's technological innovation capabilities. Fig. 1 shows the mechanism of the impact of the NEVPC policy on technological innovation for China's NEV enterprises.

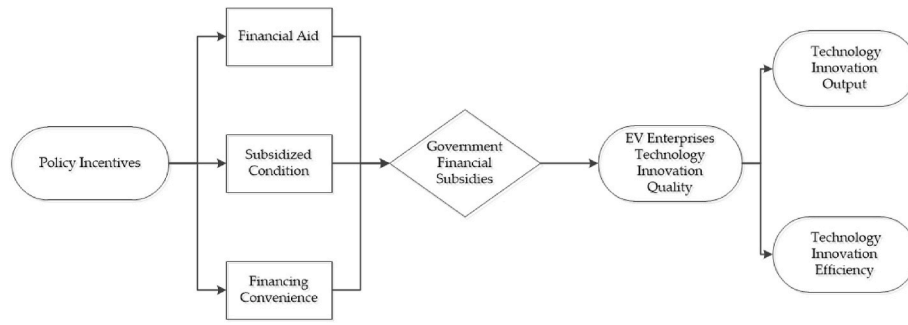


Fig. 1. The mechanism of the NEVPC policy impact on NEV enterprises technology innovation.

3. Study design and model construction

3.1. Model settings

In order to accurately identify the causal effect of policy incentives on the technological innovation of NEV enterprises based on the NEVPC policies jointly announced by several central ministries in China in 2009 and 2010, referring to Beck et al. (2010), we used a multi-period DID model for causal inferences. DID is a commonly used method that accounts for selection bias arising from time-invariant unobserved covariates, (Botosaru and Gutierrez, 2018), allowing us to relax the policy conditions and thereby making the evaluation closer to economic reality. The DID approach can also effectively alleviate an endogeneity problem in regressions. Therefore, the DID model is often used to explore the impact of an event (Tang et al., 2021). Implementing the NEVPC is a significant event in promoting a city's NEV industry, and establishing China's NEV pilot cities constitutes a quasi-natural experiment. Specifically, the selected NEV enterprises in the 25 cities are used as the treated group, and other enterprises comprise the control group. Based on this, our benchmark model (1) is constructed as follows:

$$EVTIQ_{it} = \alpha_0 + \alpha_1 DID_{it} + \rho_1 X_{it} + \mu_i + \pi_t + \varepsilon_{it} \quad (1)$$

In model (1), i represents the enterprise and t represents the year. $EVTIQ_{it}$ is the explained variable, which is a combination of the quantity of technological innovations from NEV enterprises and innovation efficiency; together they represent the technological innovation quality of the NEV enterprises. DID_{it} is the core explanatory variable and α_1 represents the implementation effect of the NEVPC. The control variable is denoted by X_{it} , π_t represents time-fixed effects and μ_i represents the individual effect. The random disturbance term is ε_{it} .

To test the mechanism by which the NEVPC policy affects technological innovation in NEV enterprises, we adopted the three-step method of the intermediary mechanism of Baron and Kenny (1986) and constructed the following recursion model based on model (1) as follows:

$$GFS_{it} = \beta_0 + \beta_1 DID_{it} + \rho_2 X_{it} + \mu_i + \pi_t + \varepsilon_{it} \quad (2)$$

$$EVTIQ_{it} = \gamma_0 + \gamma_1 DID_{it} + \gamma_2 GFS_{it} + \rho_3 X_{it} + \mu_i + \pi_t + \varepsilon_{it} \quad (3)$$

GFS_{it} is an intermediary variable representing government subsidies. Models (1), (2), and (3) together constitute a three-step method for testing the intermediary effect. If α_1 in the model (1) is significant, it indicates that the NEVPC policy incentive has a significant impact on technological innovation for NEV enterprises. In model (2), if β_1 is significant, it indicates that the NEVPC affects the government subsidies to NEV enterprises. Finally, observe the significance of γ_1 and γ_2 in the model (3), if γ_1 and γ_2 are significant, it shows that policy incentives will affect the technological innovation for NEV enterprises through a partial mediation effect of government subsidies; if γ_1 is not significant but γ_2 is significant, it indicates a complete mediation effect.

3.2. Variable setting and data description

- (1) The dependent variable is the quantity (amount) of technological innovation output (TIO). Considering that there is a time lag between a patent's application date and approval date, current patent applications better reflect the amount of innovation in a given year (Zucker and Darby, 2009). Therefore, we use the logarithm of the number of patent applications to determine TIO;
- (2) The independent variable is the DID value for a city based on its NEVPC status. The value for the year a city is named an NEV pilot city and subsequent years is 1; otherwise, the value is 0. To account for the time lag of the policy effect, we lag the two pilot phases by one year.
- (3) The mediating variable is government fiscal subsidies (GFS). We use the logarithm of the number of government subsidies awarded to NEV enterprises each year to define GFS.
- (4) Our control variables include two variables at the city level and two at the enterprise level. LNGDP represents the level of urban economic development, using the logarithm of the total GDP for each city; INDUS represents the urban industrial structure, defined as the ratio of the value of the output of a city's secondary industry to the value of the output of its tertiary industry. The cash flow ratio (CASH) represents the availability of endogenous financing for an enterprise, which will affect investment expectations and financing constraints for that enterprise, which affects its innovation activities. The ratio of net cash flow divided by the enterprise's total assets is used to define this variable; GROWTH represents the growth rate of the enterprise, represented by the year-on-year percentage growth in total assets.
- (5) Our data sources are as follows: The list of pilot cities comes from the official website of the National Energy Administration of China, and data on cities is obtained from the "China Urban Statistical Yearbook." Data on enterprises comes from the CSMAR database and the iFind database. We select NEV enterprises in other cities within the same province as the NEV pilot cities as the control group (with the exception of four municipalities and Hainan Province, as only Haikou city, which is one of the NEV pilot cities, contains any listed NEV enterprises in that province).

The data screening process is as follows: we exclude records with serious missing data, and we exclude NEV enterprises that were listed after 2006. Records for listed enterprises that are missing a minor amount of data are completed by interpolation and extrapolation, and we winsorize the data at the 1% level. After processing, we obtain balanced panel data for 196 listed NEV enterprises in China for the period from 2006 to 2020. Using data from these enterprises over this 15-year period, we examine the impact of the NEVPC on the level of technological innovation from NEV enterprises. Table 1 shows the distribution of the control and treated groups.

Table 1

Geographical distribution and number of city and enterprise in two groups of samples.

Control Group			Treated Group		
Province	Number of city	Number of enterprise	Province	Number of city	Number of enterprise
Anhui	5	7	Anhui	1	5
Beijing	/	/	Beijing	1	7
Fujian	1	1	Fujian	1	1
Guangdong	6	13	Guangdong	2	27
Guangxi	2	2	Guangxi	/	/
Hainan	/	/	Hainan	1	1
Hebei	1	1	Hebei	/	/
Henan	4	4	Henan	1	2
Heilongjiang	1	1	Heilongjiang	/	/
Hubei	1	1	Hubei	2	6
Hunan	2	2	Hunan	1	2
Jilin	1	1	Jilin	1	2
Jiangsu	7	19	Jiangsu	2	13
Jiangxi	3	3	Jiangxi	1	1
Liaoning	2	2	Liaoning	2	2
Shandong	6	9	Shandong	1	1
Shaanxi	1	1	Shaanxi	/	/
Shanghai	/	/	Shanghai	1	13
Sichuan	4	4	Sichuan	1	2
Tianjin	/	/	Tianjin	1	2
Zhejiang	7	27	Zhejiang	1	6
Chongqing	/	/	Chongqing	1	2
Total	53	98	Total	23	97

* Hohhot of Inner Mongolia Autonomous Region, and Tangshan of Hebei province have no data on listed NEV enterprises and are excluded from the treated group.

4. Empirical results and discussion

4.1. Direct effect analysis

M1 in Table 2 reports the DID model estimation results using the number of technological innovations of NEV enterprises as the dependent variable. The regression coefficient of DID is 0.326, which is significant at the 1% level, indicating a positive linear relationship between the implementation of the NEVPC and the number of technological innovations (measured by patent applications) of NEV enterprises. This indicates that NEV policy incentives can significantly increase the number of patent applications from NEV enterprises. With the policy incentives, NEV enterprises are more active in applying for patents, which indicates technological innovation. Regarding the results for the

Table 2

DID model regression results.

	M1	M2	M3
	TIO	GFS	TIO
GFS			0.0362*** (0.0113)
DID	0.3260*** (0.0850)	0.3361** (0.1444)	0.3138*** (0.0850)
LNGDP	-0.6332*** (0.1816)	1.3594*** (0.3085)	-0.6824*** (0.1819)
INDUS	0.2584*** (0.0934)	-0.1676 (0.1587)	0.2645*** (0.0933)
GROWTH	0.0002* (0.0001)	0.0014*** (0.0002)	0.0002 (0.0001)
CASH	-0.0008 (0.0019)	0.0078** (0.0032)	-0.0011 (0.0019)
Individual Fixed Effect	Y	Y	Y
Time Fixed Effect	Y	Y	Y
Observation Number	2940	2940	2940
R2	0.6910	0.4530	0.6922

* Standard errors in brackets, *, ** and *** are significant at the 10%, 5%, and 1% levels, respectively. The table below is the same.

control variables, the coefficient of LNGDP is negative and significant, indicating that the larger the city in which an NEV enterprise is located, the less conducive it is to technological innovation for these enterprises. The coefficient of INDUS is significantly positive, indicating that when the secondary industry in the city where the NEV enterprise is located is relatively important to that city, it benefits the NEV enterprise in generating technological innovation. The coefficient of GROWTH is significantly positive; indicating that the greater the enterprise's growth, the more patents it will develop.

4.2. Mediating effect analysis

M2 in Table 2 examines the relationship between the NEVPC and government subsidies. The estimation results show that implementing the policy has played a role in promoting government subsidies at the 5% significance level, with a regression coefficient for DID of 0.336. This result confirms the tasks established in the policy documents were achieved. That is, the central government supported the pilot construction in the form of government subsidies and gave financial subsidies to the NEV pilot cities. This provided economic assistance to the pilot cities' supporting infrastructure and impacted factors such as the scale effect and technological effect, which also supplement resources NEV enterprises use for innovation activities.

In the results from M3, shown in Table 2, the estimated coefficients for GFS and DID are both positive and significant at the 1% level. The estimated coefficient of government subsidies declines from 0.326 in M1 to 0.314, which indicates that government subsidies play a partial mediating role. That is, the policy incentive effect produced by the NEVPC policy not only directly affects the number of technological innovations for NEV enterprises; it also affects the number of technological innovations for enterprises through the intermediary mechanism of government subsidies.

4.3. Robustness test

4.3.1. Parallel trend test

An essential premise for the validity of the DID method is that the treated group and the control group had parallel trends prior to the policy implementation. Before the NEVPC policy is implemented, the changes in technological innovation between the NEV enterprises in the pilot cities and those in the non-pilot cities display a consistent trend. To verify the suitability of the DID model for this study, and therefore the validity of the above regression results, we use the parallel trend test to verify the changes in the trend of TIO for the two groups of enterprises.

Fig. 2 shows the results of the parallel trend test. The dotted line in the figure indicates the implementation time of the two-phase NEVPC policy. Before the policy is implemented there is no significant

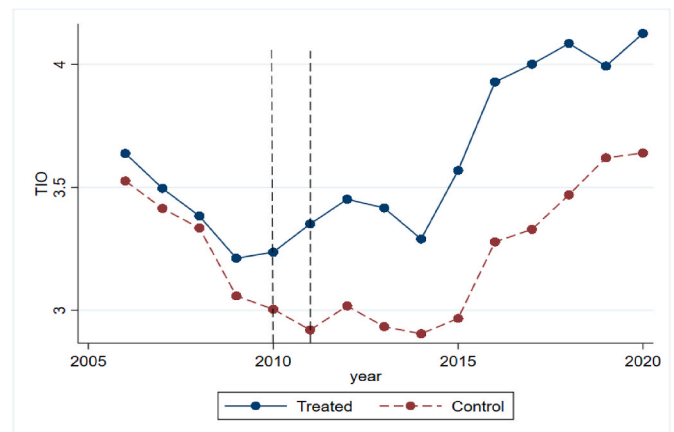


Fig. 2. Parallel trend test.

difference in the changes in trends between the treated group and the control group, and the two curves roughly fit together and move down to the right. After the two-phase policy is implemented, the two curves begin to differ significantly. Therefore, the parallel trend hypothesis is confirmed, indicating that the DID model used is applicable to this study. Furthermore, it verifies that the policy incentives created by this policy have a significant positive impact on the TIO of NEV enterprises in the pilot cities.

4.3.2. PSM-DID model test

In addition, due to the considerable heterogeneity among various enterprises in terms of scale, history, and stage of development, it is difficult to control for and ensure consistency between NEV enterprises in pilot and non-pilot cities. In ensure the two groups have characteristics that are as similar as possible, to reduce the bias caused by a non-random sample selection, we use the PSM method to match the two sample groups (Rosenbaum and Rubin, 1985). Specifically, we use the kernel matching method to match the two dependent variables, and the regression analysis is performed again using the data obtained after matching.

In Table 3, the regression coefficient for DID in M4 is 0.4578, which is significant at the 1% level. The results of the PSM-DID model are consistent with the results shown previously, which verifies that any bias caused by sample selection does not affect our conclusions.

4.3.3. The urban administrative hierarchy test

The two-phase pilot for the NEVPC policy includes four notable cities, Beijing, Shanghai, Chongqing, and Tianjin. As municipalities directly under the central government in China, these four cities are equivalent to provinces in terms of the administrative hierarchy. Given this, as a robustness test we exclude them from the sample.

M5, presented in Table 3, shows the urban administrative hierarchy test results. The regression coefficient of DID is 0.2743, which is positive and significant at the 1% level. This result verifies the robustness of the conclusions reached thus far, indicating that the sample of NEV enterprises in the four select cities do not bias the research conclusions.

4.3.4. Excluding the confounding effects of other policies

In May 2010, the “Notice on Launching the Pilot Program of Subsidies for Private Purchase of New Energy Vehicles” listed six cities—Shanghai, Changchun, Shenzhen, Hangzhou, Hefei and Beijing—as pilot cities that were given subsidies of CNY 50,000–60,000 per vehicle. In 2013, the Chinese central government launched a second demonstration project covering 88 pilot cities and a stronger promotion for NEV sales.

Table 3
Regression results of robustness test.

	M4	M5	M6	M7	M8
	PSM-TIO	TIO	TIO	TIO	TIO
DID	0.4578*** (0.1576)	0.2743*** (0.0942)	0.2639*** (0.0759)	0.1555* (0.0835)	0.1184* (0.0707)
POLICY ₂₀₁₀				0.3262*** (0.1065)	
L1.TIO					0.6577*** (0.0153)
LNGDP	0.1257 (0.3365)	−0.5447*** (0.2018)	−0.0321 (0.2234)	0.1256 (0.2285)	−0.3705** (0.1606)
INDUS	−0.3665** (0.1749)	0.2411** (0.0977)	0.1820* (0.0949)	0.1435 (0.0954)	0.0469 (0.0828)
GROWTH	−0.0002 (0.0003)	0.0003** (0.0001)	0.0002 (0.0002)	0.0002 (0.0002)	0.0002* (0.0001)
CASH	0.0090*** (0.0035)	−0.0006 (0.0020)	−0.0027 (0.0017)	−0.0030* (0.0017)	−0.0009 (0.0015)
Individual Fixed Effect	Y	Y	Y	Y	Y
Time Fixed Effect	Y	Y	Y	Y	Y
Observation Number	2555	2580	1372	1372	2744
R2	0.0982	0.6584	0.8779	0.8788	0.8225

At the same time, some local governments proposed additional incentives (e.g., financial subsidies for vehicle purchases, free license plates, parking discounts, and fee subsidies) to complement the national-level policies (Li et al., 2022; Yao et al., 2022). As these confounding and ongoing NEV policies have some overlap in terms of implementation time and region, the confounding effects are eliminated with a placebo test using a randomly generated experimental group. The coefficient from the placebo test is extracted by randomly selecting the experimental group and repeating the exercise 500 times and plotting those results in a graph compared to the actual policy effect. The results of the placebo test are shown in Fig. 3.

Fig. 3 shows that the estimated coefficients from the 500 “pseudo-policy dummy variables” are normally distributed and are mostly clustered around zero, and all of the estimated coefficients are much smaller than the coefficient estimated in M1. At the same time, most of the estimated *p*-values are above 0.100 (not significant at the 10% level), which means that the estimated results above are not obtained by chance. Therefore, the result we obtain has not been affected by other policies or random factors, which meets the expectations of the placebo test. This indicates that confounding and ongoing policies would not affect the results, proving the robustness of the main findings.

In addition, we intercepted data up to 2013 in a full-sample regression for analysis and added a policy marker variable from six pilot cities that received subsidies in 2010 to further identify the causal relationship between the NEVPC policy and technological innovations of NEV enterprises, controlling for other policy effects. The results, in M6 and M7 in Table 3, show that the estimated coefficients of the dependent variable remain highly significant after the policy marker variable is included, indicating that overlaying other policies does not affect the robustness of the main findings.

4.3.5. Robustness test for endogenous problem

Next, we consider possible endogeneity problems, such as serial correlation and inertial correlation in enterprise R&D activities, i.e., innovation performance from a previous year will be transmitted to the current year under the influence of NEVPC policy. To do so, we add the dependent variable of the previous year to the main regression for validation. The results of this regression are shown as M8 in Table 3. The policy effect of NEVPC remains significant after adding the lagged terms of the dependent variable, verifying the robustness of the core findings.

4.3.6. The staggered DID robust estimator test

The literature on multi-period DID (staggered DID) finds that in a two-way fixed effects (TWFE) framework, the core estimated

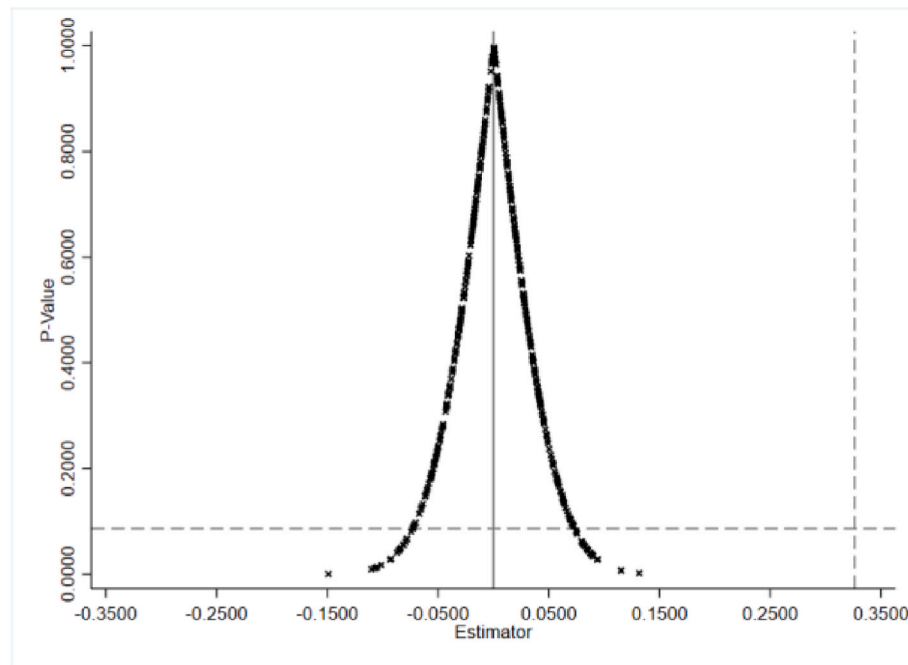


Fig. 3. Placebo test for TIO. Note: The horizontal dashed line indicates the P-value equal to 0.1. The vertical dashed line indicates the estimated coefficient of independent variable in M1.

coefficients in a multi-period DID model are a weighted average of multiple sets of underlying DID estimates and are not true treatment effects. In particular, multi-period DID estimators in the TWFE framework may have large biases when there are group and temporal heterogeneity in the treatment effects (Borusyak et al., 2021; De Chaisemartin and d'Haultfoeuille, 2020; Goodman-Bacon, 2021). Various studies have proposed different staggered DID robust estimators to correct for model estimation bias in the TWFE framework (Gardner, 2022; Sun and Abraham, 2021). Here we assess the impact of implementing the NEVPC policy on technological innovation of NEV enterprises based on the two-stage DID approach proposed by Gardner (2022).

The regression results are shown in Table 4 as M9 and M10. We see that when using the staggered DID robust estimator, the NEVPC can significantly increase the number of NEV enterprises applying for patents. The results verify the robustness of our main conclusions.

4.4. Further analysis

The results above show that government subsidies help to increase the technological innovation of NEV enterprises, measured by patent applications, via a partial intermediary role. Government subsidies are an innovation resource with clear value for NEV enterprises. The process of using these resources is bound to be accompanied by an increase in the number of technological innovations, but do these subsidies make an enterprise more efficient in terms of its ability to innovate? The technological innovation efficiency (TIE) of NEV enterprises is an issue that

deserves more attention. Do policy incentives and accompanying government subsidies make NEV enterprises in pilot and non-pilot cities differ in terms of TIE? Data Envelopment Analysis (DEA), proposed by Charnes et al. (1978), is a popular non-parametric programming method for evaluating efficiency. However, the traditional DEA model only evaluates cross-sectional data and cannot be applied to panel data. Therefore, we use the DEA window analysis model (Charnes and Cooper, 1984) to evaluate comparable efficiency values in time series data. Investment technological innovation is measured using three factors: talent input, financial investment, and material input. The number of patent applications made by NEV enterprises is selected as the TIO variable. The definitions and descriptions of the input-output indicators used in the DEA model are shown in Table 5. Wu et al. (2020) argue that a DEA window analysis achieves the best balance of reliability and stability in the efficiency measure using a window width of three. Therefore, we use the DEA window analysis model with a window width of three to evaluate TIE.

We use the kernel density estimation method to estimate TIE for NEV enterprises measured from 2006 to 2020 and present the results in Fig. 4. We can see that the kernel density of TIE for NEV enterprises in pilot and non-pilot cities is almost the same. Most of the TIE values for the two sample groups are concentrated in the range of 0.4–0.8, and the overall distribution is relatively consistent. We preliminary conclude that NEVPC policy incentives have little effect on the TIE of NEV enterprises.

We also use the *t*-test to compare the TIE of the two groups of NEV enterprises, with the results shown in Table 6. There is no significant

Table 4
Regression results of staggered DID robust estimator test.

	M9	M10
	TIO	TIO
DID	0.3722** (0.1778)	0.3344* (0.1720)
Control Variables	N	Y
Individual Fixed Effect	Y	Y
Time Fixed Effect	Y	Y

Table 5
Definitions and descriptions of the input-output indicators of the DEA model.

Indicator	Category	Description
Input	Talent input factor	Number of R&D personnel
	Financial investment factor	R&D capital investment
	Material input factor	Value of fixed assets
Output	Technology innovation output	Number of patent applications

*Taking into account the impact of past innovation investment on current and future innovation output, we use the perpetual inventory issuance to calculate the input factors.

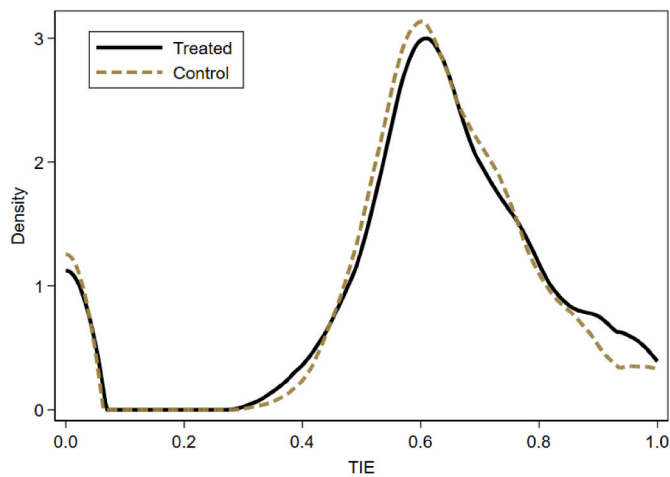


Fig. 4. Kernel density estimation result.

difference in the TIE values of the two sample groups, which further verifies that the NEVPC policy incentive does not significantly impact technological innovation efficiency for NEV enterprises.

Finally, we incorporate the calculated TIE for NEV enterprises into the regression analysis model, with the results presented in Table 7. The regression coefficient for DID in M11 is 0.0125, which is not significant. In M12, the estimated coefficients of GFS and DID are both insignificant. These regression results are consistent with the findings discussed above that the incentives and government subsidies in the NEVPC policy have little effect on the TIE of NEV enterprises.

4.5. Discussion

Using balanced panel data for 196 NEV enterprises in China consisting of 2940 observations for the period from 2006 to 2020, this study uses a staggered DID model to empirically test the impact of China's NEVPC policy on technological innovation in NEV enterprises. We found that policy incentives significantly increased the TIO of NEV enterprises in pilot cities, measured by the number of patent applications. Our results show that increasing government assistance to NEV enterprises can stimulate innovation and development in the industry.

In examining the mechanism by which the NEVPC policy impacts technological innovation in NEV enterprises, we found that implementing the policy significantly promoted increased government subsidies, verifying that goals in the policy document were realized. That is, the central government supported the pilot program by providing subsidies to NEVs in pilot cities. We also found that these government subsidies supplement innovation resources that contribute to innovation activities and can increase the TIO of NEV enterprises.

Finally, given that government subsidies provided value to NEV companies, in addition to measuring the impact of subsidies on the level of technological innovation, the efficiency of technological innovation is also worthy of attention. We use DEA window analysis to measure the TIE of NEV enterprises. Using the kernel density estimation method and *t*-test, we found that the NEVPC policy does not affect TIE for NEV enterprises. Government subsidies can help to increase the level of TIO for NEV enterprises because money is an essential and influential innovation resource, but do not improve the efficiency of technological innovations.

Table 6

T-test on the technological innovation efficiency.

Variable	Sample	Observation Number	Mean	Standard Deviation	P-value (diff<0)	P-value (diff! = 0)	P-value (diff>0)
TIE	Treated	1455	0.5872	0.2418	0.1241	0.2481	0.8759
	Control	1485	0.5870	0.2360			

5. Conclusion and policy implications

5.1. Conclusion

We collect data on the technological innovations of 196 NEV enterprises in China during the 15 years from 2006 to 2020 and empirically test the differences in TIO for such enterprises in pilot cities and non-pilot cities before and after the NEVPC policy was implemented. We verify the government subsidies in the policy, reveal the mechanism by which the policy affects technological innovation for NEV enterprises, and draw the following conclusions:

Firstly, the NEVPC policy has significantly increased the TIO of NEV enterprises in the pilot cities.

Secondly, the NEVPC policy accomplished the goal of providing certain government subsidies to the pilot cities and thus effectively boosted the TIO of NEV enterprises.

Finally, the policy effect brought by the NEVPC policy is not effective in improving the TIE of NEV enterprises.

5.2. Policy implications

These conclusions can provide important guidance for policies to support the NEV industry and promoting technological innovation of NEV enterprises. The policy implications are as follows: (1) The government should improve the effectiveness of policy support for technological innovation to further stimulate the vitality of technological innovation in the NEV industry. With the development of NEVs, awareness among consumers has increased and policy effectiveness can play a more prominent role in developing the industry. While improving technical product standards, policymakers should implement policy assistance to play a role in supporting technological innovation of NEV enterprises. (2) Policies to support the NEV industry should focus on financial subsidies but could add detailed implementation steps, specific measures, requirements, and other operational content. The extensive use of "subsidy policies" is a feature of China's industrial policies. In a period of declining subsidies and policy overlap, a policy that assists the NEV industry should be diverse in its approach. For example, policies could help to strengthen the competitive environment for NEV enterprises and establish different subsidies for different types of

Table 7

Regression results of technological innovation efficiency.

	M11 TIE	M12 TIE
GFS		−0.0006 (0.0016)
DID	0.0125 (0.0124)	0.0126 (0.0124)
LNGDP	−0.1087*** (0.0264)	−0.1079*** (0.0265)
INDUS	0.0332** (0.0136)	0.0331** (0.0136)
GROWTH	−0.0000 (0.0000)	−0.0000 (0.0000)
CASH	0.0000 (0.0003)	0.0000 (0.0003)
Individual Fixed Effect	Y	Y
Time Fixed Effect	Y	Y
Observation Number	2940	2940
R2	0.6122	0.6123

technological innovations. (3) The procuratorial agency should strengthen the application and management of government subsidies to help improve their efficiency. Our results show that government subsidies at this stage of the NEV industry have not improved TIE in NEV enterprises. The reason may be related to the lack of sound and effective rules, supervision, and accountability mechanisms for those subsidies. In the future, while providing financial support the government may want to pay more attention to the daily operation of projects funded by subsidies in the NEV industry, and improve rules regarding how subsidies are to be used. Relevant regulatory authorities should conduct periodic inspections on R&D projects of cities and enterprises that are given subsidies to prevent enterprises from misappropriating project funds for other purposes. In the final stage of a project, the mechanism for evaluating the result should be refined, and certain punitive measures should be given to the pilot cities and affiliated enterprises of NEVs that fail to meet the expected goals of the project, such as stipulating that their project application qualifications will be cancelled within a certain period in the future.

5.3. Limitations and suggestions for future studies

This study empirically investigates the policy effects of China's NEVPC policy and offers reliable conclusions that pass robustness tests. However, this study has certain limitations. First, the analysis combines TIO and TIE to measure the quality of technological innovation in NEV enterprises. Further research is needed to measure this indicator more comprehensively. For example, a comprehensive index system that includes indicators such as patents, TIE, R&D investment, and new product development could be considered.

Second, a growing number of econometric studies have focused on the limitations of using a staggered DID in a TWFE framework. In this study, the staggered DID robust estimators are measured using a two-stage DID in the robustness test. These emerging models in the DID field can be used in the baseline model for future studies.

CRediT authorship contribution statement

Zhangsheng Jiang: Conceptualization, Writing – review & editing, Review and editing, Supervision, Funding acquisition. **Chenghao Xu:** Writing – original draft, Original draft preparation, Methodology, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.enpol.2023.113527>.

References

- Ahn, J.M., Lee, W., Mortara, L., 2020. Do government R&D subsidies stimulate collaboration initiatives in private firms? *Technol. Forecast. Soc. Change* 151, 119840.
- Baron, R.M., Kenny, D.A., 1986. The moderator–mediator variable distinction in social psychological research: conceptual, strategic, and statistical considerations. *J. Pers. Soc. Psychol.* 51, 1173.
- Beck, T., Levine, R., Levkov, A., 2010. Big bad banks? The winners and losers from bank deregulation in the United States. *J. Finance* 65, 1637–1667.
- Borusyak, K., Jaravel, X., Spiess, J., 2021. Revisiting Event Study Designs: Robust and Efficient Estimation arXiv preprint arXiv:2108.12419.
- Botosaru, I., Gutierrez, F.H., 2018. Difference-in-differences when the treatment status is observed in only one period. *J. Appl. Econom.* 33, 73–90.
- Bronzini, R., Piselli, P., 2016. The impact of R&D subsidies on firm innovation. *Res. Pol.* 45, 442–457.
- Charnes, A., Cooper, W.W., 1984. Preface to topics in data envelopment analysis. *Ann. Oper. Res.* 2, 59–94.
- Charnes, A., Cooper, W.W., Rhodes, E., 1978. Measuring the efficiency of decision making units. *Eur. J. Oper. Res.* 2, 429–444.
- Choi, J., Lee, J., 2017. Repairing the R&D market failure: public R&D subsidy and the composition of private R&D. *Res. Pol.* 46, 1465–1478.
- De Chaisemartin, C., d'Haultfoeulle, X., 2020. Two-way fixed effects estimators with heterogeneous treatment effects. *Am. Econ. Rev.* 110, 2964–2996.
- Dong, F., Liu, Y., 2020. Policy evolution and effect evaluation of new-energy vehicle industry in China. *Resour. Pol.* 67, 101655.
- Gardner, J., 2022. Two-stage Differences in Differences arXiv preprint arXiv:2207.05943.
- Gass, V., Schmidt, J., Schmid, E., 2014. Analysis of alternative policy instruments to promote electric vehicles in Austria. *Renew. Energy* 61, 96–101.
- Goodman-Bacon, A., 2021. Difference-in-differences with variation in treatment timing. *J. Econom.* 225, 254–277.
- Huang, Z., Liao, G., Li, Z., 2019. Loaning scale and government subsidy for promoting green innovation. *Technol. Forecast. Soc. Change* 144, 148–156.
- Li, S., Zhu, X., Ma, Y., Zhang, F., Zhou, H., 2022. The role of government in the market for electric vehicles: evidence from China. *J. Pol. Anal. Manag.* 41, 450–485.
- Liu, L., Zhang, T., Avrin, A.-P., Wang, X., 2020a. Is China's industrial policy effective? An empirical study of the new energy vehicles industry. *Technol. Soc.* 63, 101356.
- Liu, X., Sun, X., Li, M., Zhai, Y., 2020b. The effects of demonstration projects on electric vehicle diffusion: an empirical study in China. *Energy Pol.* 139, 111322.
- Meckling, J., Nahm, J., 2019. The politics of technology bans: industrial policy competition and green goals for the auto industry. *Energy Pol.* 126, 470–479.
- Rosenbaum, P.R., Rubin, D.B., 1985. Constructing a control group using multivariate matched sampling methods that incorporate the propensity score. *Am. Statistician* 39, 33–38.
- Siddiqui, A.S., Marnay, C., Wiser, R.H., 2007. Real options valuation of US federal renewable energy research, development, demonstration, and deployment. *Energy Pol.* 35, 265–279.
- Sun, L., Abraham, S., 2021. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *J. Econom.* 225, 175–199.
- Tang, C., Xu, Y., Hao, Y., Wu, H., Xue, Y., 2021. What is the role of telecommunications infrastructure construction in green technology innovation? A firm-level analysis for China. *Energy Econ.* 103, 105576.
- Wang, S., Chen, G., Huang, D., 2021. Can the new energy vehicle pilot policy achieve green innovation and emission reduction?—a difference-in-differences analysis on the evaluation of China's new energy fiscal subsidy policy. *Sustainability* 13, 8643.
- Wu, D., Wang, Y., Qian, W., 2020. Efficiency evaluation and dynamic evolution of China's regional green economy: a method based on the Super-PEBM model and DEA window analysis. *J. Clean. Prod.* 264, 121630.
- Yao, X., Ma, S., Bai, Y., Jia, N., 2022. When are new energy vehicle incentives effective? Empirical evidence from 88 pilot cities in China. *Transport. Res. Pol. Pract.* 165, 207–224.
- Yu, F., Guo, Y., Le-Nguyen, K., Barnes, S.J., Zhang, W., 2016. The impact of government subsidies and enterprises' R&D investment: a panel data study from renewable energy in China. *Energy Pol.* 89, 106–113.
- Yuan, X., Liu, X., Zuo, J., 2015. The development of new energy vehicles for a sustainable future: a review. *Renew. Sustain. Energy Rev.* 42, 298–305.
- Zhang, L., Qin, Q., 2018. China's new energy vehicle policies: evolution, comparison and recommendation. *Transport. Res. Pol. Pract.* 110, 57–72.
- Zhao, F., Chen, K., Hao, H., Wang, S., Liu, Z., 2019a. Technology development for electric vehicles under new energy vehicle credit regulation in China: scenarios through 2030. *Clean Technol. Environ. Policy* 21, 275–289.
- Zhao, Q., Li, Z., Zhao, Z., Ma, J., 2019b. Industrial policy and innovation capability of strategic emerging industries: empirical evidence from Chinese new energy vehicle industry. *Sustainability* 11, 2785.
- Zucker, L.G., Darby, M.R., 2009. *Star Scientists, Innovation and Regional and National Immigration, Entrepreneurship and Openness*. Edward Elgar Publishing.
- Zuo, W., Li, Y., Wang, Y., 2019. Research on the optimization of new energy vehicle industry research and development subsidy about generic technology based on the three-way decisions. *J. Clean. Prod.* 212, 46–55.